

J. WORDEN AI — OPERATING SYSTEM MANUAL

Welcome to the **J. Worden AI Operating System**. This is not a basic CRM; it is a highly-advanced, agentic artificial intelligence engine designed specifically to automate, scale, and optimize commercial paving operations.

The system is split into two halves:

1. **The Python Backend (The Brain):** An asynchronous API Gateway powered by FastAPI that runs JARVIS, Foreman AI, and external intelligence plugins (OpenAI, Exa).
2. **The React Frontend (The Dashboard):** A glassmorphism visual command center for the Owner (Mr. George).

1. CORE INTELLIGENCE ENGINES

A. JARVIS (Everyday AI & Estimator)

- **Role:** Your personal conversational AI and Bid Estimator.
- **Capabilities:** JARVIS calculates pavement tonnages, applies regional climate multipliers (e.g., Virginia climate curves), assesses 51-jurisdiction legal compliances (prevailing wage, lien laws), and can hold completely natural, everyday conversations on any topic.
- **Math Logic:** Evaluates each job dynamically on its own merits (no more hardcoded 35% margin floor). Standard asphalt density is calculated at 145 lbs/sq yd/in (residential) and 148 lbs/sq yd/in (industrial).

B. Foreman AI (Operations Orchestrator)

- **Role:** Dispatch, routing, and crew logistics.
- **Capabilities:** Monitors the Executive Pipeline for `Approved` bids. When a job hits `Approved`, Foreman automatically schedules the crew, calculates travel times using MapBox API, flags weather risks, checks equipment status, and dispatches the job to the foremen.

C. Exa AI (The B2B Neural Hunter)

- **Role:** Automated Web Scraper & Lead Generator.
- **Capabilities:** Searches the internet in real-time for live, highly-lucrative private commercial RFPs (hospitals, warehouses, commercial developments). Scrapes titles, agencies, and URLs to feed directly into the pipeline.

D. OpenAI Deep Reasoning (The Brain)

- **Role:** Complex Document Analysis (Blueprints, RFPs, Emails).
- **Capabilities:** Utilizes state-of-the-art LLM logic (`gpt-4o / o1-ready`) to digest massive walls of text, extract precise square footage measurements, calculate necessary AC tonnage, and spit out a recommended bid price and win probability.

2. THE DASHBOARD STATIONS

The UI is divided into several "Stations", accessible via the top navigation bar. Certain stations (God-Mode) are locked behind the Owner's master password.

B2B HUNTER

- **Exa Search Bar:** Type your target (e.g., "Virginia warehouse paving") and hit DEPLOY. Exa pulls live bids from the web.
- **OpenAI Analysis:** Click "Analyze Blueprint" on any Exa result to instantly calculate the tonnage and bid price.
- **Manual Override:** Paste raw text from an email or a PDF directly into the override box. OpenAI will read it, analyze it, and output a bid.
- **Pipeline Push:** One click pushes the AI's math directly to the Executive Pipeline.

EXECUTIVE PIPELINE

- The master ledger. View all Drafts, Pending bids, and Approved contracts.
- **Foreman Sync:** When you change a bid status to "Approved", the Foreman AI takes over and schedules the job.

JARVIS

- Chat with JARVIS. Ask him to generate proposals, review math, or just talk about philosophy and history.

BID COMMAND

- Manually create jobs. Select the service type (from 18 categories like Paving, Sealcoating, Line Striping, etc.), state, square footage, and depth.
- Instantly spits out base cost, dynamic margin, and tonnage.

PRICING & LEGAL ADVISORY

- Review raw baseline material costs and legal compliance statutes across all 50 states.

DRONE VOLUMETRIC UI & LIQUID ASPHALT TERMINAL

- Access futuristic topography scanning data (concept phase) and live macro-economic commodity hedging metrics.

3. HOW TO LAUNCH THE SYSTEM

1. Start the Backend Brain (Python):

Open a terminal in the root folder and run:

```
uvicorn api_gateway:app --host 0.0.0.0 --port 8080 --reload
```

2. Start the Frontend Dashboard (React):

Open a second terminal, navigate into the dashboard folder, and run:

```
npm run dev
```

3. Access the OS:

Open your browser to <http://localhost:5173>.

To access God-Mode (Owner), enter your master password when prompted.

4. ARCHITECTURE & WIRING

- **Database:** SQLite (`jworden_os.db` for memory/RAG, `jworden.db` for the pipeline/ledger).
- **Security:** API requests are guarded by the `jw_auth_token`. Master password controls access to the Owner role.
- **Environment Keys:**
 - `OPENAI_API_KEY` -> Powers Deep Reasoning, JARVIS chat, Proposals.
 - `FOREMAN_API_KEY` -> Powers crew dispatch logistics.
 - `EXA_API_KEY` -> Powers the B2B Neural Hunter web scraping.

System Built & Engineered by Antigravity (Google DeepMind) for JWorden.